Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “current” to explain your model or result. Sentence starters are optional.

series: connected along one unbranched current path

current: rate of charge flow

resistance: voltage divided by current for a stated operating condition

01 Warm-up: bring back an earlier idea

Two series components have drops 3 V and 5 V. A charge of 7 C passes through each.
Find each energy transfer and the total.

Word help: charge = a property responsible for electric interactions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

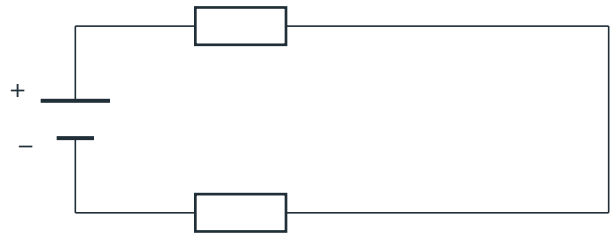
02 Worked example: explain the model

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$.
Series resistors share one unbranched current path.

Worked reasoning: Two series resistors 2 and 4 ohms on a 12 V ideal supply have total 6 ohms and current 2 A. Drops are 4 V and 8 V; they add to 12 V.

Your turn: Draw the loop and label current and both voltage drops.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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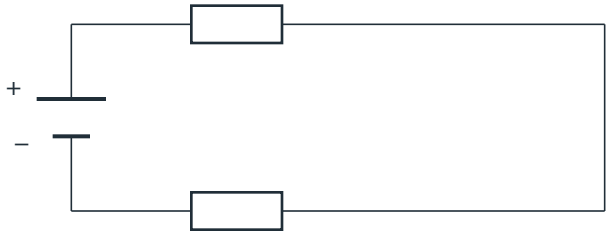
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Trace the only current path in the series diagram. If either resistor connection opens, what happens to the current?

Sentence starter: The … in my model represents … . Its label shows … .



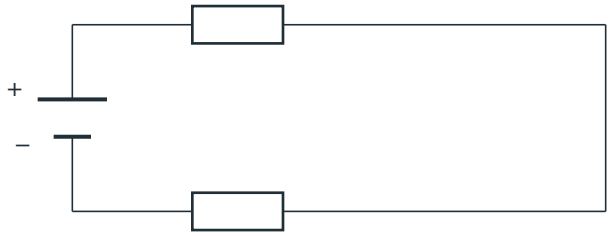
Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: 1 2 3

04 Apply the physics

A 12 V lab supply powers 10 Ω and 20 Ω resistors in series.
Label the circuit, then find total resistance, current and both voltage drops.
Use the drops to check the loop’s energy-per-charge balance.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student assigns more current to the smaller series resistor.
Explain why this would imply charge accumulating between resistors.

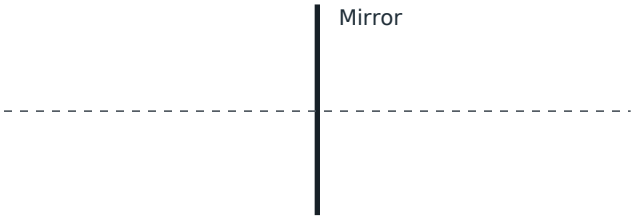
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An object moves 0.4 m toward a fixed plane mirror.
Find image movement and the change in object-image separation.
Mark the object and image positions before and after the change.

Sentence starter: My design shows … . It meets the requirement by … .



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

07 Exit check

Add a third equal resistor to two equal series resistors at the same supply voltage.
Compare new current with original current.
Label the circuit and show the current path or paths.

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Three series resistors 2, 3, R ohms draw 2 A from 20 V.
Find R and power in each resistor.
Label the circuit and show the current path or paths.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3